

Exam preparation and review

Review session

Running scenario: Thara Hospitals patient portal pilot

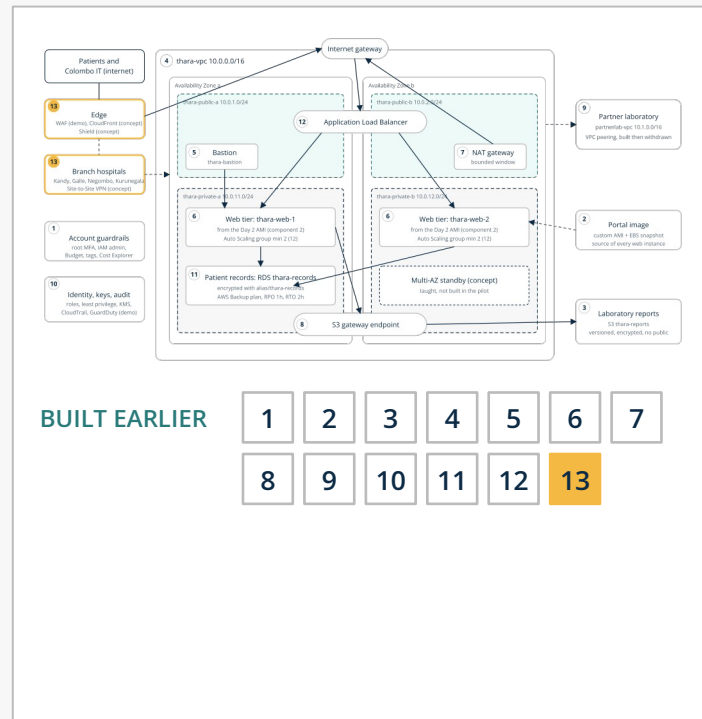
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This session

Answering in parts, a timed mock, seven ideas often confused, and the capstone.

- What a composite question asks - 20 min
- What the questions can and cannot ask - 15 min



All eight outcomes are in play

**L01**

Explain the key differences between virtualisation and containerisation.

L02

Explain cloud computing concepts.

L03

Explain high availability (HA) concepts.

L04

Describe the core features of cloud computing.

L05

Implement, scale, and manage cloud instances with storage, networking, and security features.

L06

Manage cloud data by configuring storage and backups.

L07

Explain virtual private clouds, subnets, gateways, and other cloud networking options.

L08

Describe the security features of cloud computing.

CONCEPT 1



What a composite question asks

One scenario, three parts, each asking for more than the last.

20 min

Three parts, rising demand

(a) Explain

6

What it is, how it works

Name the right mechanism and say what follows from it.

About 10 minutes.

(b) Apply

8

For this scenario

A design, a rule, a configuration or a diagram, with the scenario's own names and numbers.

About 14 minutes.

(c) Evaluate

6

Weigh it

What is right in a proposal, what is wrong, and what would change your view.

About 10 minutes.

The marks are those of your six practice questions. Your time follows the marks.

Read for the constraints before you write

- A constraint is a fact that rules answers out: a budget, a recovery target, a rule about personal data
- Every practice question held at least two, printed under "Constraints you should notice"
- When nobody prints them, underline them yourself before you plan



THARA HOSPITALS

The same three people every time: the CFO's USD 150 a month, the Head of IT's targets, and the DPO's "who can read it, and how would we know".

Time by marks, draw first, close on a trade-off

- A practice question is set for about 35 minutes: spend them in proportion to the marks
- For a network question, **draw before you describe**: VPC, subnets, route tables, gateways, the path
- Part (c) is worth as much as part (a), so do not arrive with three minutes left

End every evaluate part: I would choose X because of this constraint; I give up Y; I would change my mind if Z.

Naming is not applying

Applies it to Thara

`thara-db-sg` admits `thara-web-sg` alone, on port 3306, so a replaced web server gets in as soon as it wears that group.

Could only be about this network.

Names the mechanism

A security group is a stateful firewall that controls traffic to an instance.

True of every network in the world.

The marking workshop looks for the sentence where an answer named a mechanism and should have applied it.

CONCEPT 2

What the questions can and cannot ask

Only what the seven days taught, built or showed, and always in a scenario.

15 min

What a question can draw on, and what it cannot

- **Built in a lab:** expect to design with it, or to write a rule or a policy for it
- **Demonstrated:** expect to explain what it does and when you would choose it
- **Named only**, such as Direct Connect or CloudFront: say what it is for, and weigh it against what you built
- Nothing the module never mentioned, and never a bare list

The organisation changes from question to question. The platform does not.

Back to Day 1: an argument, not a fact

- A virtual machine has its own kernel and is isolated by the hypervisor
- A container is a group of processes on the host's kernel: a different boundary, not a thinner one
- "Which is better" has no answer until someone says for whom and for what



A small IT team, a six-month pilot and a Data Protection Officer who asks who can read the data at every layer.

Activity: timed mock, 35 minutes



1. One full question in three parts, on a Thara scenario you have not seen
2. No notes, no talking, one sheet for your drawing
3. Read for the constraints, draw, and spend your time by the marks
4. Write your student id on every sheet; your answer comes back to you for marking

Activity: marking workshop, 45 minutes



1. Mark two constructed answers against the marking guide, one part at a time
2. Agree a mark for each part with your neighbour, and the sentence that earned or lost it
3. Then mark your own answer the same way
4. Find where "names the mechanism" stopped short of "applies it"

Three questions to ask of every part

Ask, in this order

- Does it name the right mechanism?
- Does it apply it to this scenario's names, numbers and constraints?
- Does it weigh a trade-off and say what would change the choice?

Then look for

- A constraint that the answer never mentions
- A sentence that would be true of any organisation
- An evaluate part that ends without choosing

Keep your marked answer. It is the first page of your revision plan.

Activity: misconceptions clinic, 45 minutes



1. Open the misconception cards: read a belief, say whether you would have agreed, then turn the card
2. Sort each one under "I had this one" or "I did not"
3. For each one you had, explain the mechanism to your neighbour with the Thara example
4. Start your revision plan with the days your pile points to

Common mistakes: Day 3 and Day 4

COMMON MISTAKE

Security groups can deny, and network ACLs are stateful.

A group has allow rules only. A network ACL can deny, and the reply needs its own rule.

COMMON MISTAKE

A subnet named public is public.

Only a route to the internet gateway, plus a public address on the instance, makes anything reachable.

COMMON MISTAKE

Peering is transitive.

It is not. Each pair that must talk needs its own peering, or a hub.

Common mistakes: Day 6



COMMON MISTAKE

Multi-AZ improves read performance.

It does not; replicas do. A standby is a spare that waits.



COMMON MISTAKE

RPO and RTO are the same number.

One is data loss, the other is downtime.

Common mistakes: Day 2 and Day 5

COMMON MISTAKE

Reserved Instances and Savings Plans are two names for one thing.

One commits to an instance family in a region; the other to a spend per hour on compute.

COMMON MISTAKE

An explicit allow wins.

An explicit deny always wins, however specific the allow is.

Activity: capstone clinic, 20 minutes



1. As a group, tick the seven deliverables in the brief against what you hold today
2. Read the nine requirements aloud and point to where your environment meets each
3. Check the contribution statement: who built and who wrote what, signed by all members
4. Each member explains their own tier to the other two

The viva: a tier that was not yours

- Fifteen minutes for the group
- Each member is asked about a tier that was not their primary responsibility
- "Walk me through a request from a patient to the database"
- "What happens if AZ a fails" and "what would you cut if the CFO halved the ceiling"

Distinction in the viva: each member answers on a tier other than their own.

After this session



TASK

Capstone submission

The seven deliverables in the brief, as a group.

On the date announced for your cohort.

TASK

Group viva

Fifteen minutes, on a tier other than your own.

After submission.

PRACTICE

Your revision plan

By learning outcome, from your marked mock and your six practice attempts.

Questions & Discussion

Next: the capstone submission and the viva, on the dates announced for your cohort.



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